



## Background & Motivation

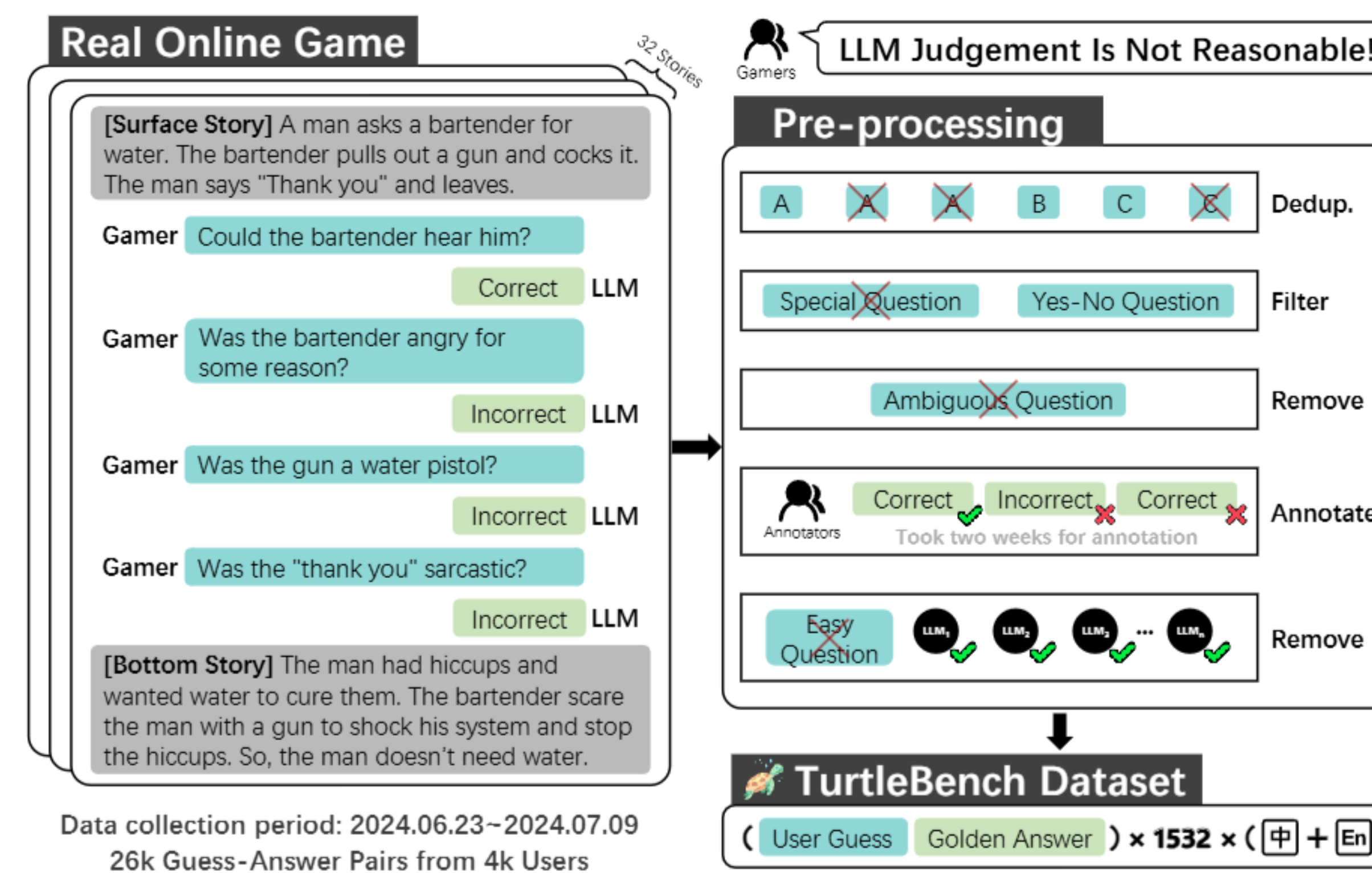
- **Limitations of Static Datasets:** Existing LLM benchmarks rely on static data, which fails to capture dynamic human-AI interactions and risks data contamination (cheating).
- **Knowledge Dependency:** Traditional benchmarks heavily rely on background knowledge, making it difficult to isolate and measure pure logical reasoning capabilities.
- **Costly & Biased Evaluations:** Dynamic evaluations using strong LLMs (e.g., GPT-4-as-a-judge) or human efforts introduce bias and are too costly for large-scale application.

## Our Solution: TurtleBench

We propose **TurtleBench**, a dynamic evaluation benchmark based on real-world "Turtle Soup" (Yes/No) puzzles.

- **Zero Background Knowledge Required:** All necessary information is self-contained within the "Surface Story" and "Bottom Story." It strictly assesses pure logical reasoning.
- **Objective & Quantifiable:** Evaluates reasoning through explicit, objective "Correct / Incorrect" ground truths, eliminating subjective scoring bias.

## Dataset Construction



We developed an online Turtle Soup puzzle platform to collect genuine human-AI interaction data:

- **Data Collection:** Gathered 26,000 guess-answer pairs from over 4,000 real users interacting with 32 distinct stories.
- **Pre-processing:** Deduplicated queries, filtered out non-Yes/No questions, and removed ambiguous guesses.
- **Annotation:** Conducted rigorous multi-turn manual annotation over two weeks to classify labels into binary "Correct" or "Incorrect".
- **Final Dataset:** 1,532 high-quality, accurately annotated bilingual (Chinese & English) evaluation pairs.

## Main Results

Table 2: Zero-Shot Evaluation Results<sup>5</sup>

| Model             | Story-Level Avg. Acc. | Overall Acc. ↑ | F1 Score |
|-------------------|-----------------------|----------------|----------|
| GPT-4o            | 88.05%                | 87.66%         | 0.8501   |
| Claude-3.5-Sonnet | 87.63%                | 87.53%         | 0.8436   |
| OpenAI o1-preview | 84.65%                | 84.40%         | 0.8071   |
| Qwen-2-72B        | 83.62%                | 82.90%         | 0.7741   |
| Moonshot-v1-8k    | 82.80%                | 82.05%         | 0.7619   |
| Llama-3.1-405B    | 82.39%                | 81.79%         | 0.8114   |
| Deepseek-V2.5     | 80.48%                | 79.77%         | 0.7368   |
| Llama-3.1-70B     | 79.44%                | 78.33%         | 0.7340   |
| OpenAI o1-mini    | 73.66%                | 73.69%         | 0.6480   |

Table 3: Two-Shot Evaluation Results

| Model             | Story-Level Avg. Acc. | Overall Acc. ↑ | F1 Score |
|-------------------|-----------------------|----------------|----------|
| Claude-3.5-Sonnet | 90.00%                | 89.49%         | 0.8729   |
| GPT-4o            | 87.89%                | 87.92%         | 0.8521   |
| Qwen-2-72B        | 85.85%                | 85.12%         | 0.8152   |
| Moonshot-v1-8k    | 84.71%                | 84.07%         | 0.8039   |
| Llama-3.1-405B    | 82.20%                | 81.72%         | 0.8061   |
| Deepseek-V2.5     | 81.70%                | 80.68%         | 0.7723   |
| Llama-3.1-70B     | 79.52%                | 79.37%         | 0.7713   |



Figure 4: Completion Token Lengths for Wrong and Right Judgments of o1-preview

## Conclusion

TurtleBench offers a reliable, dynamic, and objective framework to assess LLMs' real-world reasoning and understanding capabilities, mitigating data contamination risks.